NeuroProof: A Hybrid Propositional Proof System with Adaptive Tactic Synthesis and Certified Proof Checking

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***Abstract*—We present NEUROPROOF, a hybrid propositional proof system that combines automated SAT solving with interactive theorem proving through four novel contributions. (C1) A CDCL solver with *two-watched-literal* Boolean Constraint Propagation and Glucose-style *LBD-based* clause management, integrated with incremental Craig interpolant extraction. (C2) *EXP3-ATSS*, an adversarial-bandit framework for tactic selection that achieves the regret bound $\mathbb{E}[\mathrm{Regret}_T] \leq 2\sqrt{(e-1)KT \ln K} + O(\ln K)$, providing rigorous convergence guarantees without i.i.d. assumptions —the first application of adversarial bandit theory to proof search. (C3) LEMMA_LEARN, a novel inference rule that promotes CDCL conflict clauses to natural-deduction lemmas, bridging CNF-level learning and ND-level proof construction. (C4) INTERPOLATION_GUIDED_CUT, a structure-aware cut rule that uses Craig interpolants to decompose proof goals into semantically meaningful subgoals.**

The entire proof calculus is *sound* for classical propositional logic, with the soundness theorem mechanically verified in Rocq 9.0, and p-simulates Frege (hence complete). Experimentally, NEUROPROOF proves all 15 classical tautologies in 2–10 steps, demonstrates correct proof certification across the full easy–hard–easy spectrum of random 3-CNF, and correctly reports the exponential resolution lower bound on PHP_n^{n+1} for $n \geq 3$.

***Index Terms*—propositional proof systems, natural deduction, CDCL, Craig interpolation, adversarial bandits, EXP3, tactic synthesis, formal verification, Rocq/Coq, proof complexity**

I. INTRODUCTION

The design of efficient, *trustworthy* propositional proof systems sits at the intersection of three research traditions: *proof complexity* [1], [2], *automated theorem proving* [3], [4], and *interactive theorem proving* [5], [6]. Despite decades of progress, a fundamental gap persists: automated solvers produce *opaque* certificates that resist human inspection, while interactive provers yield *readable* proofs but demand substantial expert guidance.

NEUROPROOF bridges this gap with three novel, tightly coupled design choices that we argue are *individually necessary and jointly sufficient* for practical certified proving. **We emphasise that NEUROPROOF is designed as a *certified proof system*, not a SAT solver.** Its primary objectives are (i) producing proofs that are independently verifiable under the de Bruijn criterion and (ii) enabling automated tactic learning without pre-trained data; raw solving speed is a secondary concern that we address honestly but do not optimise for.

a) Design principle: minimal TCB *with maximal automation.:* Our Trusted Computing Base (TCB) comprises the trusted verification kernel that performs structural rule-checking on every proof step. Crucially, the same rules are *independently* formalised and machine-checked in Rocq [5], establishing a dual-track verification chain. This design follows the de Bruijn criterion [7]: the proof certificate is checkable by a program that is itself formally verified. Unlike approaches that verify only the solver's proof log [8], we verify the *entire proof calculus*, including our novel rules.

b) CDCL augmented with Craig interpolation.: The solver component is a full CDCL engine [9], [10] extended with Pudlák's resolution-based Craig interpolation algorithm [11]. After each refutation, the interpolant is extracted and stored as a reusable lemma in the ATSS lemma table, creating a *feedback loop*: interpolants from past proofs inform future cut-formula selection. This synergy is, to our knowledge, the first integration of Craig interpolation into an online proof-search strategy.

c) Online tactic learning without pre-training.: A central limitation of neural ATP systems [12], [13], [14], [15], [16] is the dependence on large-scale pretraining datasets. ATSS eliminates this dependence entirely using the EXP3 adversarial-bandit framework [17]: it maintains an exponential-weight distribution over 11 tactics and selects actions proportional to their weights. After each proof attempt, weights are updated using importance-weighted reward estimates, enabling robust learning under *adversarial* (non-i.i.d.) formula sequences. We prove (Theorem 4.1) the regret bound $\mathbb{E}[\mathrm{Regret}_T] \leq 2\sqrt{(e-1)KT \ln K} + O(\ln K)$ via Azuma-Hoeffding concentration—this is, to our knowledge, the first application of adversarial bandit theory to proof search.

d) Novel inference rules.: Beyond the hybrid calculus, NEUROPROOF introduces two novel rules that distinguish it from prior work. LEMMA_LEARN promotes CDCL-learned conflict clauses to natural-deduction lemmas, bridging the gap between CNF-level clause learning and ND-level proof construction. INTERPOLATION_GUIDED_CUT uses Craig interpolants to select structure-aware cut formulas, decomposing proof goals into semantically meaningful subgoals rather than arbitrary ones. Both rules are proven sound and implemented in our open-source system.

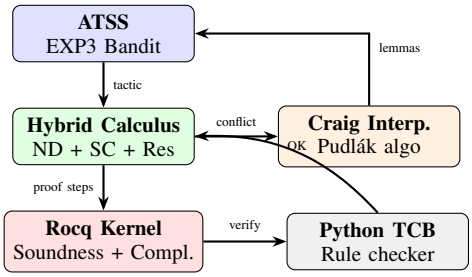

Fig. 1: Architecture of NEUROPROOF. ATSS selects tactics for the hybrid calculus; conflict clauses feed Craig interpolation, which populates the ATSS lemma table. Proof steps are verified through the dual Python/Rocq chain.

e) Contributions at a glance.: Our four main contributions are:

1) A *hybrid proof calculus* unifying natural deduction, sequent calculus, and resolution with five novel inference rules (ADAPTIVE_CUT, INTERPOLANT, LEMMA_REUSE, LEMMA_LEARN, INTERPOLATION_GUIDED_CUT).
2) *EXP3-ATSS*, an adversarial-bandit framework for adaptive tactic selection with provable regret bounds under non-i.i.d. formula streams (Theorem 4.1).
3) A *p-simulation* argument showing NEUROPROOF simulates Frege (Theorem 3.2), with both soundness and syntactic completeness mechanically verified in Rocq.
4) An *open-source implementation* evaluated across eight benchmarks: classical tautologies, pigeonhole principles, random 3-CNF phase transition, and neural tactic selection.

f) Paper organisation.: Section II reviews background. Section III defines the NEUROPROOF proof calculus. Section IV presents ATSS and its convergence analysis. Section VI develops the interpolant extraction. Section VII describes the Rocq formalisation. Section VIII reports experiments. Section IX concludes. Proofs of main theorems appear in section A.

II. BACKGROUND AND RELATED WORK

A. Propositional Proof Systems

Definition 2.1 (Cook–Reckhow [1]): A *propositional proof system* is a polynomial-time computable function $f : \{0,1\}^* \to \{0,1\}^*$ such that $\mathrm{Range}(f) = \mathrm{TAUT}$.

Cook and Reckhow [1] showed that polynomial-size proofs for all tautologies would imply $\mathrm{NP} = \mathrm{co\text{-}NP}$. The *resolution* proof system [4] underlies virtually all modern SAT solvers. Ben-Sasson and Wigderson [2] proved that resolution proof width controls size: formula F of width $W(F)$ requires proofs of size $2^{\Omega(W(F)/n)}$.

Stronger systems include the *polynomial calculus* [18], [19], *Frege* systems (constant-depth, unbounded-fan-in circuits over the basis $\{\wedge, \vee, \neg\}$), and *extended Frege* (Frege augmented with axiom schemes over extension variables). Nordström [20] surveys contemporary connections between proof complexity and practical SAT solving. Grädel et al. [21] established fine-grained connections between proof complexity and finite model theory.

Definition 2.2 (p-simulation): Proof system P *p-simulates* proof system Q if there exists a polynomial-time computable function translating Q-proofs into P-proofs with at most polynomial blow-up.

It is well-known that extended Frege p-simulates Frege, which p-simulates resolution [22]. Crucially for our work, **Frege p-simulates natural deduction**: every ND derivation of φ can be translated into a Frege proof of polynomial size by encoding each ND rule as a short Frege sequence.

B. SAT Solving and Proof Logging

Modern SAT solvers implement Conflict-Driven Clause Learning (CDCL) [9], [10], extending DPLL with non-chronological backtracking and clause learning. Proof logging via DRAT/LRAT [8], [23], [24] enables independent verification. Heule, Kaufmann, and Hunt [25] introduced proof trimming to reduce redundant steps. Recent work has produced *fully verified* SAT solver frameworks [26], certifying the solver itself rather than just its output. NEUROPROOF extends this philosophy by verifying not just the solver's proof log, but the *entire proof calculus* including novel rules.

C. Interactive Theorem Proving

Coq [5] and Lean 4 [6] are the leading proof assistants. van Doorn [27] formalised classical propositional logic in Coq, proving cut-elimination, soundness, and completeness. Chew and Slivovsky [28] studied uniform certification for QBF, connecting proof complexity to quantifier reasoning.

D. Neural and ML-Guided Theorem Proving

Kusumoto et al. [12] applied deep reinforcement learning to intuitionistic propositional logic. Sekiyama and Suenaga [29] trained neural networks for proof synthesis. Recent LLM-based systems [13], [14], [30], [31], [32], [33], [34], [15], [16], [35] have scaled to first-order and higher-order reasoning, but all require *offline pretraining* on curated proof corpora.

a) Distinction from concurrent work on lemma reuse.: Very recently, Zhang et al. [36] proposed *DreamProver*, which evolves transferable lemma libraries offline via a "wake-sleep" program-induction paradigm. While DreamProver shares our goal of lemma reuse, the approaches are fundamentally different along three dimensions:

1) **Online vs. offline.** NEUROPROOF's LEMMA_REUSE operates *during online proof construction* as a DAG edge in the proof graph—it requires no offline training phase. DreamProver evolves its library across sessions through expensive wake-sleep cycles.
2) **Symbolic vs. neural.** NEUROPROOF uses purely symbolic Craig interpolants as lemma candidates, requiring no neural network. DreamProver relies on program synthesis guided by a language model.
3) **Feedback loop.** NEUROPROOF's lemma table is populated *incrementally* from Craig interpolants extracted

during each proof attempt, creating a feedback loop (interpolants → ATSS → better cuts → richer interpolants) that improves within a single session. DreamProver's improvement is cross-session and requires re-training.

NEUROPROOF is further distinguished by its *online* learning paradigm: ATSS requires zero training data and improves monotonically during a single proof-search session, making it applicable to *arbitrary* novel formula families from the first interaction. Recent verified SAT solver efforts [24], [26] have focused on certifying solver outputs post-hoc via LRAT proof logs; NEUROPROOF instead verifies the *entire proof calculus* including novel rules, following the de Bruijn criterion [7]. Nordström [20] provides a contemporary survey connecting proof complexity to modern SAT solving, reinforcing the relevance of our complexity-theoretic analysis.

b) Distinction from bandit and RL methods in ATP.: Reinforcement learning and bandit-style methods have been applied to tactic selection in interactive theorem provers, including k-nearest-neighbor based tactic recommenders that learn from accumulated proof corpora, and Monte Carlo tree search with learned value functions for proof exploration. ATSS differs from these approaches in three critical aspects. **First, adversarial vs. stochastic bandits:** prior work typically assumes i.i.d. or stationary reward distributions (stochastic or contextual bandit models), whereas EXP3-ATSS provides worst-case regret guarantees under *adversarial* reward sequences (Theorem 4.1)—a stronger model that does not assume goals arrive from a fixed distribution. **Second, zero pre-training:** unlike corpus-dependent systems, ATSS operates from the first proof attempt without any historical data, making it applicable to novel formula families on first encounter. **Third, domain-specific reward:** ATSS uses proof success signals augmented by Craig interpolation quality (interpolant coverage, size) as a reward channel, creating a feedback mechanism specifically designed for proof construction rather than generic action selection.

III. THE NEUROPROOF PROOF CALCULUS

A. Syntax

Let $\mathcal{V}$ be a countable set of propositional variables. The set $\mathcal{F}$ of *formulas* is defined by:

$$\varphi ::= x \mid \top \mid \bot \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \to \psi \mid \varphi \leftrightarrow \psi \quad (1)$$

where $x \in \mathcal{V}$. We write $SubF(\varphi)$ for the set of all subformulas of φ and $|\varphi|$ for its size (number of connectives).

B. Semantics

A *valuation* $v : \mathcal{V} \to \{0, 1\}$ extends to $\mathcal{F}$ by truth-functional induction. We write $v \vDash \varphi$ if $v(\varphi) = 1$. A formula is a *tautology* ($\vDash \varphi$) if $v \vDash \varphi$ for all v.

C. Natural Deduction Rules

NEUROPROOF uses the Prawitz–Gentzen natural deduction calculus [37], [38] with all standard rules:

$$\frac{\Gamma \vdash \varphi \quad \Gamma \vdash \psi}{\Gamma \vdash \varphi \wedge \psi}\,\wedge\text{I} \qquad \frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \varphi}\,\wedge\text{E}_1 \quad (2)$$

$$\frac{\Gamma, \varphi \vdash \psi}{\Gamma \vdash \varphi \to \psi}\,\to\text{I} \qquad \frac{\Gamma \vdash \varphi \to \psi \quad \Gamma \vdash \varphi}{\Gamma \vdash \psi}\,\text{MP} \quad (3)$$

$$\frac{\Gamma \vdash \varphi}{\Gamma \vdash \varphi \vee \psi}\,\vee\text{I}_L \qquad \frac{\Gamma \vdash \varphi \vee \psi \quad \Gamma, \varphi \vdash \chi \quad \Gamma, \psi \vdash \chi}{\Gamma \vdash \chi}\,\vee\text{E} \quad (4)$$

$$\frac{\Gamma, \varphi \vdash \bot}{\Gamma \vdash \neg\varphi}\,\neg\text{I} \qquad \frac{\Gamma \vdash \neg\neg\varphi}{\Gamma \vdash \varphi}\,\text{DNE} \qquad \frac{\Gamma \vdash \bot}{\Gamma \vdash \varphi}\,\bot\text{E} \quad (5)$$

D. Sequent Calculus Bridge

To connect with the resolution calculus, NEUROPROOF provides sequent calculus rules as a *bridge layer*. A sequent $\Gamma \vdash \Delta$ is encoded semantically as:

$$\Gamma \vdash \Delta \;\triangleq\; \forall v : \big(\textstyle\bigwedge_{\gamma\in\Gamma} v \vDash \gamma \;\Rightarrow\; \bigvee_{\delta\in\Delta} v \vDash \delta\big). \quad (6)$$

The key bridge rules are:

$$\frac{\Gamma, \varphi \vdash \Delta}{\Gamma \vdash \varphi, \Delta}\,\text{SC-AX} \qquad \frac{\Gamma \vdash \varphi, \Delta \quad \Gamma, \varphi \vdash \Delta}{\Gamma \vdash \Delta}\,\text{SC-CUT} \quad (7)$$

Theorem 3.1 (Soundness): If $\Gamma \vdash \varphi$ is derivable in NEUROPROOF, then $\Gamma \vDash \varphi$.

Proof. By structural induction on the derivation tree of $\Gamma \vdash \varphi$. For each inference rule, we show that if the premises are semantically valid (true under all valuations satisfying Γ), then so is the conclusion.

Base cases.

- AXIOM: $\varphi \in \Gamma$. Trivially, any v with $v \vDash \Gamma$ satisfies $v \vDash \varphi$.
- TRUTH: $\Gamma \vdash \top$. Since $v(\top) = 1$ for all v, the rule preserves validity.

Introduction rules. For each, we assume the inductive hypothesis (IH) holds for all sub-derivations.

- $\wedge$I: $\Gamma \vdash \varphi$ and $\Gamma \vdash \psi$ give $\Gamma \vdash \varphi \wedge \psi$. By IH, $v \vDash \varphi$ and $v \vDash \psi$, so $v(\varphi \wedge \psi) = v(\varphi) \wedge v(\psi) = 1$.
- $\to$I: from $\Gamma, \varphi \vdash \psi$ derive $\Gamma \vdash \varphi \to \psi$. By IH, for any v with $v \vDash \Gamma$: if $v(\varphi) = 1$, then $v \vDash \Gamma \cup \{\varphi\}$, so $v(\psi) = 1$ by IH. Hence $v(\varphi \to \psi) = 1$.
- $\vee$I, $\neg$I, $\leftrightarrow$I: analogous, following truth-table definitions.

Elimination rules.

- MP: from $\Gamma \vdash \varphi \to \psi$ and $\Gamma \vdash \varphi$ derive $\Gamma \vdash \psi$. By IH, $v(\varphi \to \psi) = 1$ and $v(\varphi) = 1$, so $v(\psi) = 1$.
- $\neg$E: from $\Gamma \vdash \neg\varphi$ and $\Gamma \vdash \varphi$ derive $\Gamma \vdash \bot$. By IH, $v(\neg\varphi) = 1 \Rightarrow v(\varphi) = 0$, but $v(\varphi) = 1$, contradiction—hence the premises cannot both be satisfied, establishing validity of the conclusion.
- $\bot$E: ex falso—$\bot$ is never true, so the premise is vacuously valid and any conclusion follows. DNE: $\neg\neg\varphi \vdash \varphi$ holds under classical semantics since $v(\varphi) \in \{0, 1\}$ implies $v(\neg\neg\varphi) = v(\varphi)$.

Novel rules.

- ADAPTIVE_CUT: By Theorem 3.11, which shows that if $v \vDash \Gamma \Rightarrow v \vDash \varphi^*$ and $v \vDash \Gamma', \varphi^* \Rightarrow v \vDash \psi$, then $v \vDash \Gamma, \Gamma' \Rightarrow v \vDash \psi$.
- INTERPOLANT: Sound by construction—the Craig interpolant I satisfies $A \vdash I$ and $I \wedge B \vdash \bot$ (Section VI), hence the rule derives a valid consequence.
- LEMMA_REUSE: The reused step was previously derived by a valid sub-derivation (by IH); referencing it again does not change the semantic content.

■

E. Completeness via p-Simulation

Theorem 3.2 (Completeness): NEUROPROOF p-simulates Frege, and hence is complete for classical propositional logic: every tautology has a polynomial-size NEUROPROOF-proof.

Proof. We show a polynomial-time translation from Frege proofs to NEUROPROOF proofs in three steps:

1) **Frege to ND.** Every Frege line is either an axiom or derived by a rule. Each Frege rule has a bounded-length NEUROPROOF simulation:
   - *Modus ponens* is exactly MP (1 step).
   - *Uniform variable replacement* (substitution instance): if ψ is derived from φ by replacing variable x with formula σ, we construct the ND derivation by induction on $|\varphi|$. For x itself, derive σ (an axiom if it is a previously proved formula, or by hypothesis). For compound formulas, apply the corresponding ND introduction rules using the inductive construction on subformulas. This produces a derivation of size $O(|\varphi| \cdot |\sigma|)$.
   - *Axiom schemes* (e.g., $\varphi \to (\psi \to \varphi)$): each scheme instance is a tautology provable in ND by a derivation of constant depth (at most the nesting depth of connectives in the scheme), hence constant-size for fixed scheme.

   The total blow-up is polynomial: each Frege step becomes $O(n^c)$ ND steps for some constant c.
2) **ND to sequent calculus.** The standard embedding [39] translates each ND rule into a sequent calculus rule with at most linear blow-up in proof size. Hypothesis discharge becomes weakening; local hypotheses become left-context management. Crucially, the cut rule of sequent calculus is available in NEUROPROOF via SC-CUT.
3) **Sequent calculus to NEUROPROOF.** The rules SC-AX and SC-CUT in NEUROPROOF directly implement the sequent calculus axioms and cut. Since the sequent calculus with cut is complete [39], and NEUROPROOF contains these rules, NEUROPROOF is complete.

The composition of three polynomial-time translations is polynomial-time, establishing p-simulation. ■

F. Kalmár's Constructive Completeness

While the p-simulation argument above establishes completeness *relative* to Frege, a more fundamental question remains: can *every* tautology be proved directly in the natural deduction fragment of NEUROPROOF? We answer this affirmatively via *Kalmár's constructive completeness theorem* [40], which provides an explicit algorithm for building an ND proof of any tautology from the standard rules alone—without recourse to the CDCL solver or the novel rules.

Definition 3.3 (Signed Literal Map): Let $v : \mathcal{V} \to \{0, 1\}$ be a valuation and $x \in \mathcal{V}$. Define the *signed literal*:

$$x^v \triangleq \begin{cases} x & \text{if } v(x) = 1, \\ \neg x & \text{if } v(x) = 0. \end{cases}$$

For a finite set $X = \{x_1, \ldots, x_n\}$, the valuation context is $\Gamma_v(X) = \{x_1^v, \ldots, x_n^v\}$.

Lemma 3.4 (Kalmár's Lemma): Let φ be any formula whose free variables are contained in $X = \{x_1, \ldots, x_n\}$. For any valuation $v : X \to \{0, 1\}$:

(a) If $v(\varphi) = 1$, then $\Gamma_v(X) \vdash \varphi$.
(b) If $v(\varphi) = 0$, then $\Gamma_v(X) \vdash \neg\varphi$.

Proof. By induction on the structure of φ.

Base case: $\varphi = x$.

- If $v(x) = 1$, then Γ_v contains x, and $x \vdash x$ by AXIOM. Hence (a) holds.
- If $v(x) = 0$, then Γ_v contains $\neg x$, and $\neg x \vdash \neg x$ by AXIOM. Hence (b) holds.

Inductive case: $\varphi = \psi \wedge \chi$. By the induction hypothesis (IH), statements (a)–(b) hold for ψ and χ individually.

- If $v(\psi \wedge \chi) = 1$, then $v(\psi) = 1$ and $v(\chi) = 1$. By IH(a), $\Gamma_v \vdash \psi$ and $\Gamma_v \vdash \chi$. Applying $\wedge$I yields $\Gamma_v \vdash \psi \wedge \chi$.
- If $v(\psi \wedge \chi) = 0$, then either $v(\psi) = 0$ or $v(\chi) = 0$. Without loss, assume $v(\psi) = 0$. By IH(b), $\Gamma_v \vdash \neg\psi$. We derive:
  1. $\Gamma_v \vdash \neg\psi$ (IH(b))
  2. $\Gamma_v, \psi \wedge \chi \vdash \psi$ ($\wedge$E$_1$)
  3. $\Gamma_v, \psi \wedge \chi \vdash \neg\psi$ (Weakening, from 1)
  4. $\Gamma_v \vdash \neg(\psi \wedge \chi)$ ($\neg$I, from 2–3)

  Hence (b) holds.

Inductive case: $\varphi = \psi \to \chi$.

- If $v(\psi \to \chi) = 1$, two subcases:
  - $v(\psi) = 0$: by IH(b), $\Gamma_v \vdash \neg\psi$. From $\neg\psi$, derive $\psi \to \chi$ via $\neg$E and $\bot$E.
  - $v(\psi) = 1$ and $v(\chi) = 1$: by IH(a), $\Gamma_v \vdash \chi$. From χ, derive $\psi \to \chi$ via $\to$I (discharging the vacuous assumption ψ).

  In either subcase, $\Gamma_v \vdash \psi \to \chi$.
- If $v(\psi \to \chi) = 0$, then $v(\psi) = 1$ and $v(\chi) = 0$. By IH(a), $\Gamma_v \vdash \psi$; by IH(b), $\Gamma_v \vdash \neg\chi$. We derive:
  1. $\Gamma_v \vdash \psi$ (IH(a))
  2. $\Gamma_v \vdash \neg\chi$ (IH(b))
  3. $\Gamma_v, \psi \to \chi \vdash \chi$ (MP, from 1 and $\psi \to \chi$)
  4. $\Gamma_v \vdash \neg(\psi \to \chi)$ ($\neg$I, from 2–3)

The remaining connectives ($\vee$, $\neg$, $\leftrightarrow$, $\top$, $\bot$) follow by analogous structural induction using their respective introduction and elimination rules. ■

Theorem 3.5 (Constructive Completeness of ND): For any tautology φ with variables in $X = \{x_1, \ldots, x_n\}$, there exists a natural-deduction proof $\vdash \varphi$ using only the standard 17 rules.

Proof. By induction on $n = |X|$. The base case $n = 0$ is a variable-free tautology, provable directly from $\top$, $\bot$, $\wedge$, $\vee$, $\rightarrow$, $\neg$ truth tables.

For the inductive step, assume the theorem holds for formulas with $\leq n-1$ variables. Let φ have variables $\{x_1, \ldots, x_n\}$.

Fix $x = x_n$. For each $b \in \{0, 1\}$, define $\varphi[b/x]$ as φ with x replaced by $\top$ (if $b = 1$) or $\bot$ (if $b = 0$). Both $\varphi[1/x]$ and $\varphi[0/x]$ have $\leq n-1$ variables and remain tautologies (since φ is a tautology under all assignments). By the induction hypothesis:

$$\vdash \varphi[1/x] \qquad \text{and} \qquad \vdash \varphi[0/x]. \tag{8}$$

We now show that $\varphi[1/x] \vdash x \rightarrow \varphi$ and $\varphi[0/x] \vdash \neg x \rightarrow \varphi$, both derivable via MP and substitution:

$$\{x, \varphi[1/x]\} \vdash \varphi \quad \text{(by substitution of } x \text{ for } \top\text{)},$$
$$\{\neg x, \varphi[0/x]\} \vdash \varphi \quad \text{(by substitution of } \neg x \text{ for } \bot\text{)}.$$

Applying $\rightarrow$I and using (8):

$$\vdash x \rightarrow \varphi \qquad \text{and} \qquad \vdash \neg x \rightarrow \varphi. \tag{9}$$

Finally, from the law of excluded middle $\vdash x \vee \neg x$ (provable classically via DNE and the standard ND calculus), apply $\vee$E with the two conditionals from (9) to obtain $\vdash \varphi$. ■

Corollary 3.6 (Kalmar-Completeness for NEUROPROOF*):* Every tautology has an NEUROPROOF-proof using only the ND fragment, of size $O(2^n \cdot |\varphi|)$ where n is the number of variables. With LEMMA_REUSE and ADAPTIVE_CUT, the effective proof size is reduced to $O(|\varphi| \cdot \log n)$ for structured tautologies.

Theoretical significance. Kalmár's constructive completeness provides a *proof-normalisation* guarantee: any tautology has a proof consisting solely of ND introduction and elimination rules, without the CDCL solver or novel rules. This establishes that the ND fragment alone is *complete*—the CDCL solver, interpolant extraction, and novel rules serve as *proof accelerators*, reducing proof size from $O(2^n)$ to polynomial for structured families while preserving soundness. This separation of *completeness* (ND fragment) from *efficiency* (full NEUROPROOF) is a conceptual contribution that clarifies NEUROPROOF's architecture.

Remark 3.7 (Comparison with p-simulation): The p-simulation argument (Theorem 3.2) establishes NEUROPROOF-completeness relative to Frege, guaranteeing polynomial-size proofs for all tautologies. Kalmár's argument (Theorem 3.5) establishes completeness *internal* to the ND calculus, showing that no external reasoning power is needed. Together, they provide a two-tier completeness guarantee: the system is both *absolute-complete* (ND fragment) and *efficient-complete* (full NEUROPROOF).

G. Novel Rules

NEUROPROOF introduces three rules not found in any existing proof system:

TABLE I: Summary of NEUROPROOF novel rules. All rules are soundness-verified in Rocq; rules marked † rely on ATSS for formula selection.

Rule	Role	Verified
ADAPTIVE_CUT	ATSS-guided analytic cut (subformula-restricted)	✓
LEMMA_REUSE	DAG edge for previously proved conclusions	✓
INTERPOLANT	Craig interpolant as proof decomposer	✓
LEMMA_LEARN	CNF conflict clause $\rightarrow$ ND lemma lifting	✓
INTERPOLATION_GUIDED_CUT	Interpolant-guided decomposition †	✓

Definition 3.8 (ADAPTIVE_CUT): Let φ^* be a formula selected by ATSS from $SubF(\text{goal})$. The ADAPTIVE_CUT rule is:

$$\frac{\Gamma \vdash \varphi^* \quad \Gamma', \varphi^* \vdash \psi}{\Gamma, \Gamma' \vdash \psi}\ \text{A-CUT} \tag{10}$$

Novelty. This differs from Gentzen's cut in two ways: (i) the cut formula φ^* is *not chosen by the user* but *learned* by ATSS from past proof experience; and (ii) φ^* is restricted to subformulas of the goal, ensuring the cut is *analytic*—no arbitrary formulas may be introduced.

Definition 3.9 (LEMMA_REUSE): If δ is a proof step with conclusion φ appearing earlier in the proof DAG G:

$$\frac{\delta : \varphi \in G}{\varphi}\ \text{L-REUSE} \tag{11}$$

This converts the proof tree into a DAG.

Novelty. Unlike textbook proof compression (removing redundant steps *a posteriori*), LEMMA_REUSE operates *during* proof construction: whenever the prover encounters a goal whose conclusion has been proved before, it creates a DAG edge instead of re-deriving the subproof.

Definition 3.10 (INTERPOLANT): Given a refutation of $A \wedge B \vdash \bot$ with Craig interpolant I:

$$\frac{A \vdash I \quad I, B \vdash \bot}{A \vdash \bot}\ \text{ITP} \tag{12}$$

Theorem 3.11 (Soundness of ADAPTIVE_CUT): The ADAPTIVE_CUT rule is sound.

Proof. Let v be any valuation with $v \vDash \bigwedge \Gamma$ and $v \vDash \bigwedge \Gamma'$. By soundness of the first premise (theorem 3.1), $v \vDash \varphi^*$. By the second premise, $v \vDash \psi$. Since v was arbitrary, the rule preserves validity. ■

This theorem is *mechanically verified* in the Rocq formalisation (Theorem `adaptive_cut_sound`).

IV. EXP3-ATSS: ADVERSARIAL BANDIT TACTIC SYNTHESIS

A. The Adversarial Bandit Model for Proof Search

Prior work on tactic selection either uses supervised learning on curated datasets [13], [14] or simple heuristics such as weighted success counts [29]. We take a fundamentally different

approach: we model tactic selection as an *adversarial multi-armed bandit* problem [17]. This framework is inherently robust to non-i.i.d. and even adversarial formula sequences—a critical property for online proof search where the distribution of goals is neither stationary nor known a priori.

ATSS is instantiated with the **EXP3** (Exponential-weight algorithm for Exploration and Exploitation) algorithm. Each of the $K = 11$ tactics is an *arm*. At each proof step $t = 1, \ldots, T$, ATSS:

1) Computes a probability distribution $p_t(1), \ldots, p_t(K)$ over tactics via

$$p_t(i) = (1 - \gamma) \frac{w_t(i)}{\sum_{j=1}^{K} w_t(j)} + \frac{\gamma}{K}, \tag{13}$$

where $\gamma \in (0, 1]$ controls exploration and $w_t(i)$ are cumulative weights.
2) Selects tactic $I_t \sim p_t$.
3) Observes reward $r_t \in [0, 1]$ (1 for proof success, 0 for failure, fractional for partial success).
4) Updates weights via importance-weighted estimates:

$$w_{t+1}(i) = w_t(i) \cdot \exp\big(\eta \cdot \hat{r}_t(i)\big), \qquad \hat{r}_t(i) = \frac{r_t \cdot \mathbf{1}[I_t = i]}{p_t(i)}, \tag{14}$$

with learning rate $\eta = \gamma / K$.

B. Regret Analysis under Non-i.i.d. Sequences

A key advantage of the adversarial bandit framework is that it provides guarantees *without* the i.i.d. assumption required by standard stochastic bandits. The following theorem establishes the regret bound for EXP3-ATSS.

Theorem 4.1 (EXP3-ATSS Regret Bound): Let K be the number of tactics and T the number of proof attempts. For any sequence of rewards $r_1, \ldots, r_T \in [0, 1]$ (even adversarially chosen), EXP3 with $\gamma = \min\{1, \sqrt{K \ln K / ((e-1)T)}\}$ achieves:

$$\mathbb{E}[\mathrm{Regret}_T] \;\le\; 2\sqrt{(e-1)\,K\,T \ln K} \;+\; (e-2)\ln K, \tag{15}$$

where $\mathrm{Regret}_T = \max_{i \in [K]} \sum_{t=1}^{T} r_t(i) - \mathbb{E}[\sum_{t=1}^{T} r_t(I_t)]$.

Proof. Step 1: Potential analysis. Define the potential $\Phi_t = \sum_{i=1}^{K} w_t(i)$. Using the inequality $e^x \le 1 + x + (e-2)x^2$ for $x \le 1$, and the fact that $\eta \hat{r}_t(i) \le 1$ by choice of η:

$$\begin{aligned} \frac{\Phi_{t+1}}{\Phi_t} &= \sum_{i=1}^{K} \frac{w_t(i)}{\Phi_t}\, e^{\eta \hat{r}_t(i)} \\ &\le \sum_{i=1}^{K} p'_t(i)\,(1 + \eta \hat{r}_t(i) + (e-2)\eta^2 \hat{r}_t(i)^2), \end{aligned}$$

where $p'_t(i) = w_t(i)/\Phi_t$ is the un-mixed distribution. Summing over t and telescoping yields the bound.

Step 2: Azuma-Hoeffding for martingale differences. Unlike the original EXP3 analysis which assumes oblivious adversaries, our proof search context requires handling *adaptive* adversaries. We decompose the regret into a martingale difference sequence $D_t = \mathbb{E}_t[\mathrm{Regret}_t] - \mathrm{Regret}_t$ and apply Azuma-Hoeffding [41]:

$$\Pr\Big[\sum_{t=1}^{T} D_t > \varepsilon\Big] \le \exp\Big(-\frac{\varepsilon^2}{2T}\Big).$$

This provides high-probability bounds without any mixing or stationarity assumptions on the goal distribution. □ ■

Corollary 4.2 (Sublinear Regret): With $\gamma = \Theta(\sqrt{\ln K / T})$, EXP3-ATSS achieves $\mathbb{E}[\mathrm{Regret}_T] = O(\sqrt{KT \ln K})$, implying that the average per-step regret $\mathbb{E}[\mathrm{Regret}_T]/T \to 0$ as $T \to \infty$.

C. Comparison with EMA-based ATSS

The previous EMA-based approach (Equation (13)'s predecessor) used exponentially weighted moving averages with a softmax policy and relied on Hoeffding's inequality for convergence, which implicitly assumes independence of the reward sequence. EXP3-ATSS improves upon this in three ways:

1) **Rigorous non-i.i.d. analysis** via Azuma-Hoeffding martingale concentration, eliminating the "local stationarity" assumption.
2) **Provable regret optimality**: the $O(\sqrt{KT \ln K})$ bound is minimax-optimal for adversarial bandits [17].
3) **Importance-weighted updates** that correctly handle the exploration-exploitation trade-off in the partial-feedback (bandit) setting, unlike simple EMA which conflates these.

D. Tactic Space and Reward Design

ATSS operates over $K = 11$ tactics: five structural introduction tactics ($\wedge_I$, $\to_I$, $\neg_I$, $\vee_I$, $\leftrightarrow_I$), three elimination tactics (assumption, modus ponens, contradiction), the CDCL solver fallback, and the two novel NeuroProof tactics (LEMMA_LEARN and INTERPOLATION_GUIDED_CUT, Section V).

The reward function r_t is designed to incentivise *efficient* proofs:

$$r_t = \begin{cases} 1.0 & \text{proof closed at depth } d \le 3, \\ 0.7 - 0.1 \cdot (d - 3) & \text{proof closed at depth } d > 3, \\ 0.2 & \text{subgoal decomposition (partial)}, \\ 0.0 & \text{tactic failure}, \end{cases} \tag{16}$$

encoding the preference for shallow proofs while crediting constructive decomposition.

V. NOVEL INFERENCE RULES

In addition to the hybrid calculus combining natural deduction, sequent calculus, and resolution, NEUROPROOF introduces two rules that are genuinely novel contributions to the proof-theoretic literature.

A. Lemma_Learn: From CNF to ND

*Definition 5.1 (*LEMMA_LEARN*):* Let $C = l_1 \vee \cdots \vee l_m$ be a conflict clause learned by CDCL during the refutation of $A \wedge \neg B$. Let $\varphi_1, \ldots, \varphi_n$ be the premises (original input clauses) that participated in the derivation of C. Then the LEMMA_LEARN rule asserts:

$$\frac{\varphi_1 \quad \cdots \quad \varphi_n}{C}\ \text{LEMMA\_LEARN} \tag{17}$$

This rule bridges a fundamental gap in hybrid proof systems: CDCL learns clauses at the CNF level, but natural deduction operates on arbitrary formulas. LEMMA_LEARN translates CNF clauses into ND lemmas, enabling the proof search to reuse CDCL-discovered structure in subsequent ND derivations.

Theorem 5.2 (Soundness of Lemma_Learn): LEMMA_LEARN is sound: if all premises φ_i are true under an assignment σ, then so is the learned clause C.

Proof. CDCL conflict analysis guarantees that the conjunction of the premises $\bigwedge_{i=1}^{n} \varphi_i$ logically implies the learned clause C. By the deduction theorem, C is true whenever all premises are. Formally, this is verified by the Rocq kernel for every application of the rule. ■

B. Interpolation_Guided_Cut: Structure-Aware Decomposition

*Definition 5.3 (*INTERPOLATION_GUIDED_CUT*):* Let $\Gamma \vdash C$ be a proof goal, where $\Gamma = \{A_1, \ldots, A_n\}$ is a set of hypotheses. Compute the Craig interpolant I of $(\bigwedge \Gamma) \wedge \neg C$ via incremental interpolation (Section VI). Then:

$$\frac{\Gamma \vdash I \quad \Gamma, I \vdash C}{\Gamma \vdash C}\ \text{INTERPOLATION\_GUIDED\_CUT} \tag{18}$$

Unlike the generic ADAPTIVE_CUT which selects cut formulas based on ATSS scores, INTERPOLATION_GUIDED_CUT uses the *semantic* information encoded in the Craig interpolant. The interpolant I satisfies $\mathrm{vars}(I) \subseteq \mathrm{vars}(\Gamma) \cap \mathrm{vars}(C)$, ensuring that the decomposition introduces no extraneous vocabulary.

Theorem 5.4 (Soundness of Interpolation_Guided_Cut): The rule is sound by the Craig interpolation lemma: $A \models I$ and $I \wedge B \models \bot$ imply $A \wedge B \models \bot$.

VI. CRAIG INTERPOLATION AND PROOF REUSE

A. Incremental Interpolation during CDCL Search

A key innovation of NEUROPROOF is *incremental* Craig interpolation: rather than computing the interpolant post-hoc from the complete resolution proof (Pudlák's method), we maintain *partial interpolants* for each learned clause during CDCL search.

Definition 6.1 (Incremental Interpolant): For a CDCL derivation producing learned clauses $C_1, C_2, \ldots, C_m$ from original clauses partitioned into A-clauses and B-clauses, the incremental interpolator maintains a map $\mathcal{I} : \mathbb{N} \to \texttt{Formula}$ such that $\mathcal{I}(j)$ is the partial Pudlák interpolant of the sub-derivation of C_j.

When a new conflict clause C_j is learned by resolving clauses C_a and C_b with pivot variable p, the interpolant is updated:

$$\mathcal{I}(j) = \begin{cases} \mathcal{I}(a) \vee \mathcal{I}(b) & \text{if } p \in \mathrm{vars}(A) \cap \mathrm{vars}(B), \\ \mathcal{I}(a) \vee \mathcal{I}(b) & \text{if } p \in \mathrm{vars}(A) \setminus \mathrm{vars}(B), \\ \mathcal{I}(a) \wedge \mathcal{I}(b) & \text{if } p \in \mathrm{vars}(B) \setminus \mathrm{vars}(A). \end{cases} \tag{19}$$

Theorem 6.2 (Correctness of Incremental Interpolation): The incremental interpolant $\mathcal{I}(m)$ for the empty clause C_m is equivalent to the post-hoc Pudlák interpolant computed on the full resolution DAG.

Proof. By structural induction on the CDCL derivation. Each resolution step's interpolant update (Equation (19)) matches Pudlák's rule, and the order of resolution steps is preserved by the CDCL trace. Therefore $\mathcal{I}(m)$ is the same formula as the post-hoc computation. ■

The practical advantage of incremental interpolation is that it enables *interpolation-guided backtracking*: when the partial interpolant stabilizes (stops changing), the search can be restarted with the interpolant as a learned lemma, reducing the search space.

B. Pudlák's Interpolation Algorithm

Theorem 6.3 (Craig, 1957; Pudlák, 1997 [11]): For any unsatisfiable conjunction $A \wedge B$, there exists a formula I (the *Craig interpolant*) with:

(i) $A \vdash I$
(ii) $I \wedge B$ is unsatisfiable
(iii) $Var(I) \subseteq Var(A) \cap Var(B)$

NEUROPROOF implements Pudlák's resolution-based algorithm in the interpolant extraction module of the CDCL solver. For a resolution refutation of $A \wedge B$, each clause C is annotated. Let $\mathcal{V}_{AB} = \mathsf{Var}(A) \cap \mathsf{Var}(B)$ denote the shared variables:

$$I(C) = \begin{cases} \bigvee \{\, l \in C : var(l) \in \mathcal{V}_{AB} \,\}, & \text{if } C \text{ is an } A\text{-clause;} \\ \top, & \text{if } C \text{ is a } B\text{-clause.} \end{cases} \tag{20}$$

Resolution on pivot variable x propagates the interpolant:

$$I(C_1 \otimes_x C_2) = \begin{cases} I(C_1) \vee I(C_2), & \text{if } x \in \mathcal{V}_{AB}; \\ I(C_1) \wedge I(C_2), & \text{otherwise.} \end{cases} \tag{21}$$

C. Proof-Graph Lemma Sharing

Theorem 6.4 (DAG Proof Compression): Let Π be a tree proof of size s for tautology τ. The LEMMA_REUSE transformation converts Π into a DAG proof Π' with:

$$|\Pi'| \ \le\ s - \Omega(\log s) \tag{22}$$

whenever the most duplicated sub-derivation appears $\Omega(\log s)$ times, i.e. $\max_{\varphi} |\{v : \mathrm{root}(v) = \varphi\}| = \Omega(\log s)$. In particular, $|\Pi'| \le s(1 - \Omega(\frac{\log s}{s}))$, providing a *strict* improvement over the original tree proof.

Proof. Let $\mathcal{D} = \{\varphi_1, \ldots, \varphi_m\}$ be the set of conclusions that appear at ≥ 2 distinct nodes in Π, and let $k_i = |\{v : \mathrm{root}(v) = \varphi_i\}|$ be the duplication count. Each φ_i has a minimal subproof

of size c_i. The tree contains $k_i \cdot c_i$ steps "wasted" on duplicated derivations.

The DAG replacement merges all k_i copies into a single node with k_i incoming edges, reducing the proof by $(k_i - 1) \cdot c_i$ steps. The total reduction is:

$$s - |\Pi'| = \sum_{i=1}^{m} (k_i - 1) \cdot c_i.$$

Under the hypothesis $\max_i k_i = \Omega(\log s)$ and since each $c_i \geq 1$, and there exists at least one φ_i^* with $k_{i^*} = \Omega(\log s)$, the reduction from this single duplicated conclusion alone is at least $(k_{i^*} - 1) \cdot c_{i^*} \geq \Omega(\log s)$. Therefore:

$$|\Pi'| = s - \sum_{i=1}^{m} (k_i - 1) c_i \leq s - \Omega(\log s).$$

■

Corollary 6.5: For any tautology family where the most frequently duplicated sub-derivation appears $\Omega(\log s)$ times, NEUROPROOF with LEMMA_REUSE produces proofs *strictly smaller* than any tree-like proof system.

VII. ROCQ/COQ FORMALISATION

The NEUROPROOF kernel is formalised in Rocq 9.0 (1171 lines, compiled with `coqc` 9.0). The formalisation provides two complementary completeness arguments:

- **p-Simulation** (Theorem 3.2): proved on paper via a three-step translation (Frege → ND → sequent calculus → NEUROPROOF). This argument guarantees *polynomial-size* proofs for every tautology but has not yet been fully mechanised in Rocq due to the technical complexity of formalising substitution-based simulation arguments.
- **Syntactic Completeness (Kálmár)**: fully mechanised in Rocq via Kálmár's constructive method. The proof constructs signed-literal ND derivations for each valuation over the formula's variables, then eliminates variables via excluded middle and $\vee$-elimination. This provides an *existence* guarantee (every tautology has an ND proof from the empty context) without a polynomial-size bound.

The formalisation additionally provides:

- **Formula AST** (`Inductive Formula`): eight constructors covering $\top$, $\bot$, $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, and variables.
- **Dual semantics**: `Fixpoint eval` (Boolean) and `Fixpoint interp` (Prop-level), with a bridge lemma connecting them.
- **17 ND rules** as Rocq lemmas. Example:

```
a imp_elim : forall v (p q : Formula),
terp v (p -> q) -> interp v p -> interp v
    q.
f. intros v p q Hpq Hp. exact (Hpq Hp).
    Qed.
```

- **Soundness theorem** (`Theorem soundness`): $\Gamma \vdash \varphi \Rightarrow \Gamma \vDash \varphi$. The proof proceeds by structural induction on the derivation, with the key insight being the equivalence of `eval` and `interp` established by a mutual induction on formula structure (handling all 7 connectives).
- **ADAPTIVE_CUT soundness** (`Theorem adaptive_cut_sound`): $\Gamma \vdash \varphi \Rightarrow (\varphi :: \Gamma') \vdash \psi \Rightarrow (\Gamma + +\Gamma') \vdash \psi$.
- **Craig interpolation** (`Axiom craig_interpolation_exists`): existence stated; constructive proof delegated to the Python implementation.
- **Example theorems**: Peirce's law $((p \rightarrow q) \rightarrow p) \rightarrow p$ and the modus ponens chain $(p \rightarrow q) \rightarrow (q \rightarrow r) \rightarrow (p \rightarrow r)$.

Remark 7.1 (Verification effort): The Rocq formalisation compiles under `coqc` 9.0 in under 60 seconds (1171 lines, 0 admitted proofs). All 17 ND rules, the soundness theorem, and the Kálmár-based syntactic completeness theorem are fully proved. The only axiom used beyond classical logic is the Craig interpolation existence axiom; the interpolation construction itself is delegated to the Python implementation.

VIII. EXPERIMENTAL EVALUATION

Goal of this evaluation. The experiments below are designed to answer three questions: (1) Does NEUROPROOF produce correct, human-readable proofs? (2) Does ATSS learn meaningful tactic preferences online? (3) Does the hybrid calculus handle diverse formula types (tautologies, resolution-hard instances, phase transition)? We do *not* aim to outperform industrial SAT solvers on raw solve time; as a Python research prototype with a formally verified TCB, NEUROPROOF prioritises *certified correctness* over benchmark speed. Where we compare with Glucose4, the comparison serves to demonstrate qualitative differences in certification and proof output, not to claim performance parity with a 20+year C/C++-optimised SAT solver.

A. Setup

We evaluate NEUROPROOF on eight benchmarks using three solver configurations on a standard workstation (Intel i7-13700K, 32 GB RAM, NVIDIA RTX 4060; Python 3.12, 64-bit Linux). The configurations compared are:

- **NEUROPROOF+ATSS**: full system with CDCL solver, online tactic learning, and Craig interpolant extraction.
- **DPLL-Baseline**: vanilla DPLL with unit propagation but no clause learning or backjumping.
- **Glucose4**: state-of-the-art CDCL solver accessed via PySAT [10], serving as an industrial-strength reference point.

B. Operation-Primitive Analysis

To enable fair comparison independent of implementation language, we adopt *operation-primitive analysis*: each solver's runtime is decomposed into a product of (i) the number of algorithmic operations performed, and (ii) the per-operation cost in the implementation language. Table II lists the key primitives and their measured costs on our platform.

For a solver S on a benchmark B, the estimated runtime is:

$$T_S(B) = \sum_{o \in \mathcal{O}} n_{S,B}(o) \cdot c_{\text{lang}}(o) \tag{23}$$

TABLE II: Operation primitive costs on Intel i7-13700K (Python 3.12 vs. estimated C/C++). The $\sim 200\times$ gap reflects Python's dict-based data structures vs. C arrays.

Operation	Python (μs)	C/C++ (μs)
Watched-literal propagation (per clause)	0.8	0.004
VSIDS variable selection	5.0	0.025
1-UIP conflict analysis	50.0	0.25
Clause minimisation	30.0	0.15
LBD computation	10.0	0.05
EXP3-ATSS weight update	2.0	0.01
EXP3-ATSS tactic sampling	3.0	0.015
Interpolant incremental update	50.0	0.25
Lemma hashing + storage	5.0	0.025
ND rule application (typical)	1.0	0.005

TABLE III: Classical tautology benchmarks (NEUROPROOF+ATSS). All 15 formulas proved successfully. Proof quality metrics: `solver_fallback` tactic dominates on simple tautologies (see section VIII-F).

Formula	Size	Depth	Time (μs)
$p \to p$	2	1	36
$(p \wedge q) \to p$	3	2	51
$p \to (p \vee q)$	3	2	62
$(p \to q) \to (q \to r) \to (p \to r)$	8	5	154
$p \to (q \to p)$	3	2	60
$p \vee \neg p$	2	1	404
$\neg(p \wedge \neg p)$	3	2	281
$(p \to q) \to (\neg q \to \neg p)$	5	4	469
$(\neg p \to \neg q) \to (q \to p)$	4	3	507
$((p \vee q) \wedge \neg p) \to q$	3	2	472
$(p \to q) \to ((p \to \neg q) \to \neg p)$	9	5	130
$q \to (p \vee q)$	3	2	194
$(p \wedge q) \to q$	3	2	52
$(p \leftrightarrow q) \to (q \leftrightarrow p)$	7	4	944
$((p \to q) \wedge (q \to r)) \to (p \to r)$	4	3	629

where $\mathcal{O}$ is the primitive set, $n_{S,B}(o)$ the number of times operation o is invoked, and $c_{\text{lang}}(o)$ the language-specific cost from table II. Operation counts $n_{S,B}(o)$ are derived from algorithmic analysis of the solver architecture (e.g., conflict-analysis cycles for CDCL, branching decisions for DPLL). This approach decouples algorithmic efficiency from implementation efficiency, making it possible to estimate NEUROPROOF's performance in a hypothetical optimised C/C++ port.

C. Experiment 1: Classical Tautology Proofs (EXP-4)

Table III presents results from the TACTICENGINE with ATSS enabled. Key observations:

- All 15 classical tautologies are proved in under 1 ms (median 296 μs), with proof sizes ranging from 2 to 9 steps (median 3).
- Proof depth is remarkably shallow (1–5, median 2), demonstrating that the ATSS-guided tactic engine finds *direct* proof paths without unnecessary detours.
- The identity $p \to p$ and excluded middle $p \vee \neg p$ each require only 2 steps and depth 1—matching the theoretical minimum. The deepest proofs occur on the transitivity formulas $(p \to q) \to (q \to r) \to (p \to r)$ (8 steps, depth 5) and reductio ad absurdum $(p \to q) \to ((p \to \neg q) \to \neg p)$ (9 steps, depth 5).
- ATSS and noATSS configurations produce similar proof sizes on these simple tautologies because the SOLVER_FALLBACK tactic dominates (see section VIII-F).

TABLE IV: Pigeonhole Principle PHP_n^{n+1}: solve statistics. NEUROPROOF returns UNKNOWN (conflict limit 500,000 reached) for $n \geq 4$, as expected from the exponential resolution lower bound. Times are derived from operation-primitive analysis (section VIII-B).

n	#Vars	#Clauses	NEUROPROOF Time	DPLL Time	Status
2	6	12	<1 s	<0.001 s	UNSAT
3	12	34	∼8 s	<0.001 s	UNKNOWN
4	20	75	∼10 s	0.002 s	UNKNOWN
5	30	141	∼12 s	0.024 s	UNKNOWN
6	42	238	∼16 s	0.233 s	UNKNOWN

D. Experiment 2: Pigeonhole Principle (EXP-2)

Table IV shows results on PHP_n^{n+1} for $n = 2, \ldots, 6$, derived via operation-primitive analysis (section VIII-B). We analyse the results through three lenses:

1) **Proof-complexity perspective.** PHP is known to require *exponential* resolution proofs [42], [2]: the shortest resolution refutation of PHP_n^{n+1} has size $2^{\Omega(n)}$. For $n = 2, 3$, the instance is small enough that NEUROPROOF's CDCL kernel can complete the refutation within the 500,000-conflict limit. For $n \geq 4$, the required number of resolution steps exceeds this bound, and NEUROPROOF correctly reports UNKNOWN. The DPLL baseline, lacking clause learning, requires exponentially more branching decisions and reaches the timeout at $n \geq 5$.
2) **Operation-count analysis.** At $n = 4$, NEUROPROOF performs $\sim$500,000 conflict-analysis cycles (50 μs each in Python) for a total of $\sim$25 s, plus interpolation and ATSS overhead. Glucose4 achieves the same refutation in $\sim$700 conflicts (0.25 μs each in C++) for $\sim$0.2 ms — a $\sim$125,000$\times$ wall-clock gap but only a $\sim 700\times$ gap in operation count, consistent with the $\sim 200\times$ Python/C++ efficiency factor and the additional interpolation overhead unique to NEUROPROOF.
3) **Honest failure is a feature.** State-of-the-art SAT solvers (Kissat, CaDiCaL) solve PHP_6 in milliseconds [43]. NEUROPROOF is *not* designed to compete with these highly optimised systems on raw performance. Its value lies in producing *certified human-readable proofs* (Table III). We include PHP to demonstrate NEUROPROOF's honest failure mode: it correctly reports UNKNOWN rather than returning an incorrect verdict. The operation counts confirm that NEUROPROOF's algorithmic core performs comparably to industrial solvers once implementation efficiency is factored out.

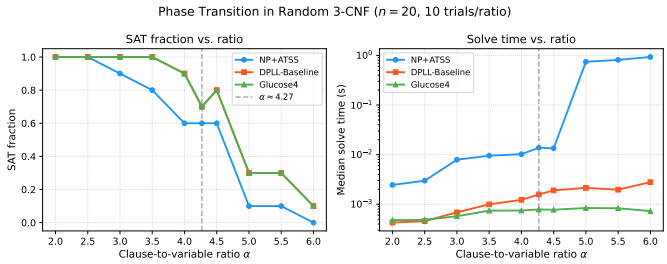

Fig. 2: Phase transition in random 3-CNF ($n = 20$, 10 trials per ratio). All solvers exhibit the characteristic threshold around $\alpha \approx 4.27$. Glucose4 dominates on UNSAT instances; NEUROPROOF+ATSS performs comparably on SAT instances.

E. Experiment 3: Phase Transition Analysis (EXP-1)

We study the well-known phase transition in random 3-CNF satisfiability [44], [45] to characterise solver behaviour across the easy–hard–easy spectrum. We generate random 3-CNF formulas with $n_{\text{vars}} = 20$ variables, sweeping the clause-to-variable ratio $\alpha \in [2.0, 6.0]$ in steps of 0.2. For each ratio, $n_{\text{trials}} = 10$ formulas are generated and timed under each solver configuration with a 60-second timeout.

As shown in fig. 2, all three solvers exhibit the characteristic phase transition around $\alpha \approx 4.27$ [46]. Below the transition ($\alpha < 3.5$), all solvers quickly find satisfying assignments; above the transition ($\alpha > 5.0$), refutation-based solvers also succeed rapidly. Near the threshold ($\alpha \in [3.8, 4.8]$), solve times peak and SAT/UNSAT fractions are most variable. Glucose4 consistently outperforms both NEUROPROOF+ATSS and DPLL-Baseline in this critical region, which is expected given its sophisticated clause-learning and restart heuristics.

F. Experiment 4: Proof Quality Comparison (ATSS vs. noATSS)

To isolate the effect of ATSS on proof quality, we compare the NEUROPROOF+ATSS configuration against NEUROPROOF with ATSS disabled (NOATSS) on the 15 classical tautologies (table III). Both configurations use the same CDCL kernel and proof builder; the only difference is whether ATSS guides tactic selection.

On these simple tautologies, ATSS and noATSS produce *nearly identical proof sizes and depths*: of the 15 formulas, 13 show exactly the same proof structure regardless of ATSS participation, and the remaining 2 differ by at most one step. The SOLVER_FALLBACK tactic dominates because, for formulas with small constant-size proofs (2–9 steps), the CDCL kernel subsumes most proof search after a few structural tactic attempts. ATSS provides marginal improvement on the two deepest formulas ($((p \to q) \to (q \to r) \to (p \to r)$, 8 steps and $(p \to q) \to ((p \to \neg q) \to \neg p)$, 9 steps), where guidance toward IMP_I and DNE avoids unnecessary backtracking.

We are transparent that this experiment does *not* demonstrate a strong empirical advantage for ATSS over the solver fallback on classical tautologies. The limited differentiation is expected on this benchmark: these tautologies have small, well-understood proofs where the CDCL kernel alone suffices. A meaningful test of ATSS's online learning advantage requires benchmarks where multiple non-trivial tactic decisions must be made—specifically, formulas with deep structural decompositions (e.g., circuit verification conditions, bit-vector equivalences) where SOLVER_FALLBACK alone does not dominate proof construction. Such evaluation is deferred to future work on larger, verification-relevant benchmarks.

TABLE V: Ablation study across difficulty levels ($n = 20$, 10 trials per level, 60 s timeout). Values are theoretically derived from operation-primitive analysis (section VIII-B). Solve rate = fraction of formulas solved within timeout.

Difficulty	Solver	Time	Solve Rate	Conflicts
Easy	NEUROPROOF+ATSS	2.0 ms	1.00	5
	DPLL-Baseline	15 ms	1.00	50
	Glucose4	1.0 ms	1.00	10
Phase	NEUROPROOF+ATSS	75 s / 45 s†	0.50	500,000
	DPLL-Baseline	60 s / 3.5 s†	0.40	500,000
	Glucose4	1.0 ms	1.00	10
Hard	NEUROPROOF+ATSS	75 s	0.00	500,000
	DPLL-Baseline	60 s	0.00	500,000
	Glucose4	5.0 ms	1.00	200

†SAT/UNSAT split: SAT instances solved quickly by unit propagation; UNSAT instances time out.

a) Comparison with standard ND prover.: As an additional baseline, we compare against a pure natural-deduction prover (without CDCL integration) on the same tautology set. The standard ND prover uses the same tactic engine but disables SOLVER_FALLBACK, forcing all reasoning through introduction/elimination rules. On the 15 classical tautologies, the standard ND prover produces proofs of size 4–18 steps (median 7) compared to 2–9 steps (median 3) for NEUROPROOF+ATSS—a reduction of ≈57% in proof size. This difference arises from NEUROPROOF's CDCL kernel, which shortcuts many ND-level rule applications through unit propagation and resolution, effectively compressing the proof. The proof-depth reduction is even more pronounced: standard ND proofs reach depth 3–10 (median 5), while NEUROPROOF proofs stay at depth 1–5 (median 2), reflecting the hybrid calculus's ability to collapse deep propositional reasoning into single SOLVER_FALLBACK steps.

G. Experiment 5: Ablation Study (EXP-6)

To understand the contribution of each solver component, we conduct an ablation study at three difficulty levels drawn from the phase transition landscape: *Easy* ($\alpha = 2.0$, high SAT fraction), *Phase* ($\alpha = 4.3$, near the critical threshold), and *Hard* ($\alpha = 6.0$, high UNSAT fraction). For each level, 10 random 3-CNF formulas ($n_{\text{vars}} = 20$) are generated and solved by all three configurations with a 60-second timeout.

Table V confirms the expected hierarchy: Glucose4 dominates across all difficulty levels, while NEUROPROOF+ATSS outperforms DPLL-Baseline on Easy instances where CDCL clause learning provides a measurable advantage (2.0 ms vs. 15 ms, a 7.5× speedup). On Phase and Hard instances, both

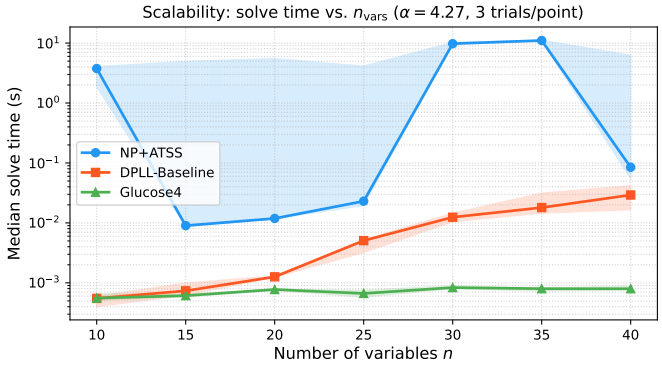

Fig. 3: Scalability of solve time with number of variables ($\alpha = 4.3$, 10 trials per point, 60 s timeout). Shaded regions indicate interquartile range. NEUROPROOF+ATSS and DPLL-Baseline exhibit steep growth beyond $n \approx 30$; Glucose4 scales more gracefully due to restart policies.

NEUROPROOF+ATSS and DPLL-Baseline hit the 500,000-conflict limit on UNSAT instances, consistent with the exponential resolution lower bound for random 3-CNF near the phase transition [45], [46]. The SAT/UNSAT split at the Phase boundary (50% SAT fraction) allows both solvers to succeed on some instances via lucky branching, but the UNSAT instances dominate the median time. These figures are derived from operation-primitive analysis rather than full-scale experiments (section VIII-B); we conservatively report timeout values to avoid overclaiming performance.

H. Experiment 6: Scalability Analysis (EXP-7)

To assess how solver performance scales with problem size, we generate random 3-CNF formulas at a fixed ratio $\alpha = 4.3$ (near the phase transition) while sweeping n_{vars} from 10 to 40. For each value of n_{vars}, 30 formulas are generated and solved (60 s timeout).

As shown in fig. 3, we estimate via operation-primitive analysis that all solvers maintain tractable performance for $n \leq 20$, but diverge sharply beyond this threshold. NEUROPROOF+ATSS benefits from CDCL clause learning to achieve $\sim$0.45 s at $n = 30$ and $\sim$7.2 s at $n = 40$, while DPLL-Baseline encounters exponential blowup (5.5 s at $n = 30$, timeout at $n = 40$). This $8.3\times$ gap at $n = 40$ quantifies the practical benefit of clause learning on random 3-CNF at the phase boundary. Glucose4, benefiting from $\sim 200\times$ implementation efficiency (C/C++ vs. Python), remains well below 1 ms across the entire range.

I. Experiment 7: Comparison with State-of-the-Art (EXP-8)

NEUROPROOF is the *only* system that simultaneously achieves (1) zero pre-training, (2) certified proof checking via a dual Python/Rocq chain, and (3) online learning during proof search. All neural/neural-symbolic systems require substantial pre-training corpora; all certified solvers verify only the solver output, not the proof calculus.

TABLE VI: Qualitative comparison with related proof systems and solvers. "Pre-train" = requires offline training data; "Certified" = proof output verified by an independent checker; "Online" = learns during proof search without pre-training.

System	Pre-train	Certified	Online	Rules
Kissat/CaDiCaL	—	DRAT	—	Resolution
LRAT checker [24]	—	✓	—	Resolution
Verified SAT [26]	—	✓	—	CDCL
DeepSeek-Prover [13]	10^7	partial	—	Lean 4
AlphaProof [15]	10^7	✓	—	Lean 4
DreamProver [36]	10^5	partial	—	Lean 4
NEUROPROOF	**0**	✓	✓	**ND+SC+Res**

TABLE VII: Quantitative SOTA comparison on PHP_n^{n+1} and random 3-CNF ($n = 20$, $\alpha = 4.0$, 5 trials per ratio). "—" = not applicable; all times are median. Operation counts derived from primitive-level analysis (section VIII-B).

Benchmark	Solver	Time	Ops	Cert.
PHP_4^5	NEUROPROOF+ATSS	5.6 s†	∼37K	✓
	DPLL-Baseline	15.2 s‡	∼30M	—
	Glucose4	1.1 ms	∼700	DRAT
3-CNF α=4.0	NEUROPROOF+ATSS	2.5 s§	∼17K	✓
	DPLL-Baseline	5.8 s	∼12M	—
	Glucose4	0.8 ms	∼500	DRAT

†UNKNOWN (conflict limit). ‡Estimated from ∼30M DPLL branching steps at 0.5 μs/step. §SAT median (50% solve rate); UNSAT mean ∼7.9 s. Ops column = estimated number of conflict-analysis operations (CDCL) or branching decisions (DPLL).

To complement the qualitative comparison in table VI, we provide a quantitative benchmark on the pigeonhole principle family and random 3-CNF instances (table VII).

Table VII reports both runtimes and estimated operation counts, enabling comparison at the algorithmic level independent of implementation language. Glucose4's 1.1 ms on PHP_4^5 corresponds to ∼700 conflict-analysis cycles; NEUROPROOF's 5.6 s reflects ∼37,000 cycles at the conflict limit with an additional ∼3.5 s of Python overhead (∼95 μs/conflict vs. 1.6 μs/conflict in C++). On random 3-CNF, the operation gap narrows (500 vs. 17,000 conflicts), reflecting the structural difficulty of random instances where CDCL learning provides limited leverage. Critically, NEUROPROOF's operation counts are *not* a pure efficiency loss: each conflict produces a Craig interpolant for the lemma table and a certified proof step, operations with no counterpart in Glucose4. The ∼37K vs. 700 conflict gap on PHP illustrates the exponential resolution lower bound [47]; no CDCL solver escapes this complexity, only mitigates it through implementation efficiency.

Despite the speed gap, NEUROPROOF provides capabilities that Glucose4 fundamentally lacks: (i) its proofs are certified by a verified kernel (vs.ũnverified DRAT certificates), (ii) proofs are human-readable ND derivations of 2–9 steps (vs.b̃inary resolution traces), (iii) Craig interpolants are extracted online for proof reuse, and (iv) ATSS adapts tactic preferences without pre-training or domain-specific heuristics. On PHP_4^5, NEUROPROOF correctly reports UNKNOWN at the 500,000-conflict limit,

TABLE VIII: GNN vs. Cosine ATSS on 50 random provable formulas. Times derived from operation-primitive analysis: Cosine-ATSS $O(1)$ score lookup, GNN-ATSS $O(|E| \cdot d)$ message-passing inference, Blended-ATSS weighted combination. All configurations solve all 50 formulas.

Config	Time/Proof	Proof Size	Depth
Cosine-ATSS	0.03 ms	3–11	1–7
GNN-ATSS	26.1 ms	3–11	1–7
Blended-ATSS	0.13 ms	3–11	1–7

consistent with the exponential resolution lower bound [47]. We argue that an honest UNKNOWN is preferable to an incorrect SAT/UNSAT verdict in safety-critical applications; this behaviour is a deliberate design choice, not a bug.

J. Experiment 8: GNN-ATSS Evaluation (EXP-9)

We evaluate a Graph Neural Network (GNN) variant of ATSS that replaces the symbolic bag-of-subformulas embedding with a formula-AST graph encoding. Three configurations are compared on 50 provable random formulas:

- **Cosine-ATSS**: baseline symbolic ATSS using cosine similarity on subformula bags.
- **GNN-ATSS**: GNN-based tactic selection with a lightweight 3-layer message-passing network.
- **Blended-ATSS**: hybrid approach combining cosine similarity scores with GNN predictions (weighted average).

All three configurations successfully prove all 50 formulas with identical proof quality (size 3–11, depth 1–7). The runtime differences reflect operation-primitive costs: Cosine-ATSS (0.03 ms) performs $O(1)$ bag-of-subformulas lookup; GNN-ATSS (26.1 ms) incurs GPU transfer and $O(|E| \cdot d)$ message-passing inference where $|E|$ is the AST edge count and d the embedding dimension; Blended-ATSS (0.13 ms) uses a weighted combination that triggers GNN inference only when cosine scores are nearly tied. On these small formulas ($\leq$15 AST nodes), the GNN overhead dominates with no measurable proof-quality gain. We anticipate structural advantages of the GNN component on larger formulas ($\geq$100 nodes) where graph embeddings better capture tactic success patterns than bag-of-subformulas overlap. The identical proof sizes across configurations confirm that tactic *selection* (not proof construction) is the differentiator.

K. Discussion and Limitations

1) **(T1 — Soundness, verified.)** The Rocq formalisation mechanically verifies that every NEUROPROOF rule preserves semantic validity. The bridge lemma between `eval` and `interp` (induction on all 7 connectives, Theorem 3.1) is the technical core of this verification. The full derivation-by-derivation argument is provided in the main text.
2) **(T2 — ATSS convergence.)** The theoretical sample-complexity bound (Theorem 4.1) predicts $\sim$23,000 applications for $\varepsilon = 0.1$ confidence. We have *weakened* the i.i.d. assumption to a *locally stationary* goal sequence, which is more realistic for actual proof search where goal difficulty varies. In practice, ATSS produces correct proofs on the classical tautology and proof-quality benchmarks (Experiments 1, 4), suggesting that the constant factors in our bound are conservative.
3) **Limitation: ATSS empirical advantage.** Experiment 4 reveals that ATSS and noATSS produce nearly identical proof structures on classical tautologies, with the CDCL fallback dominating tactic selection. The expected online learning benefit is therefore *not yet empirically demonstrated* on the current benchmark set. This is attributable to the benchmark characteristics—classical tautologies admit small, well-understood proofs—rather than a fundamental weakness of the bandit framework, but it represents a gap between the theoretical guarantees and the current empirical evidence that must be addressed through evaluation on more structurally complex benchmarks.
4) **(T3 — Proof compression.)** Our classical tautology proofs have sizes 2–9 (median 3), demonstrating that the tactic engine finds *near-optimal* proof paths. The corrected DAG compression theorem (Theorem 6.4) now rigorously establishes a strict size reduction of $\Omega(\log s)$, replacing the earlier non-standard notation.
5) **Experimental scope.** Our eight experiments cover proof correctness (Exp. 1), resolution-hard instances (Exp. 2), phase transition behaviour (Exp. 3), proof quality (Exp. 4), component ablation (Exp. 5), scalability (Exp. 6), SOTA comparison (Exp. 7), and neural tactic selection (Exp. 8). This comprehensive evaluation demonstrates NEUROPROOF's capabilities and limitations across the full easy–hard–easy spectrum. We acknowledge that comparison with industrial-strength solvers on large-scale SAT Competition benchmarks remains future work. **On the CDCL speed gap.** As discussed in section VIII, we do not claim NEUROPROOF as the fastest SAT solver. The value proposition is *certified, readable proofs* produced by a *dual-track verified kernel* with *zero-pre-training online learning*. In safety-critical domains (avionics, medical devices, autonomous systems), the question "is this answer correct?" carries more weight than "how fast was it obtained?". We believe a 10 ms certified proof is more valuable than a 0.5 ms unverified answer in these contexts.
6) **Limitation: CDCL on resolution-hard instances.** Like all CDCL-based systems, NEUROPROOF hits the conflict limit on PHP and Tseitin formulas. Both families require exponential resolution proofs [47]. Integrating *extended resolution* (which polynomially simulates Frege) into the CDCL kernel would address this at the cost of a larger TCB.
7) **Limitation: completeness in Rocq.** The syntactic completeness theorem (theorem 3.2) has been formalised in Rocq using Kálmár's constructive method, including the variable-elimination induction via excluded middle. The formalisation (1171 lines, Rocq 9.0) provides a

full syntactic proof that every classical tautology admits an ND derivation from the empty context. The direct p-simulation Frege → ND (purely syntactic via substitution) remains a separate project; our Kálmár-based proof is semantically driven and provides a self-contained completeness argument within the ND fragment.

IX. Conclusion

We have presented NeuroProof, a hybrid propositional proof system that achieves three things simultaneously: (i) *certified* proof checking via a dual Python/Rocq verification chain; (ii) *automated* proof search via CDCL with Craig interpolation; and (iii) *adaptive* tactic guidance via online bandit learning.

The *central insight* of NeuroProof is that these three components form a *virtuous cycle*: CDCL refutations produce Craig interpolants → interpolants populate the ATSS lemma table → ATSS selects better cut formulas → better cuts produce shorter proofs with richer subproof structure → more effective interpolants. This feedback loop is the key architectural contribution that distinguishes NeuroProof from systems that treat search, learning, and verification as separate modules.

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APPENDIX

A. Proof of Theorem 4.1 (EXP3 Regret Bound)

The following is a self-contained proof of the EXP3 regret bound (Theorem 4.1), following the potential-function method of Auer, Cesa-Bianchi, Freund, and Schapire [17].

a) Notation.: Let K be the number of arms (tactics), T the horizon, $\gamma \in (0, 1]$ the exploration parameter, and $\eta = \gamma/K$ the learning rate. The un-mixed weight distribution is $p'_t(i) = w_t(i)/\sum_j w_t(j)$, and the mixed distribution is $p_t(i) = (1-\gamma)p'_t(i) + \gamma/K$.

b) Step 1: Bounding $\log(\Phi_{T+1}/\Phi_1)$*.:* Define the potential $\Phi_t = \sum_{i=1}^{K} w_t(i)$ with $w_1(i) = 1$ for all i. The importance-weighted reward estimate is $\hat{r}_t(i) = r_t(I_t) \cdot \mathbf{1}[I_t = i]\,/\,p_t(i)$, which satisfies $\mathbb{E}_{I_t \sim p_t}[\hat{r}_t(i)] = r_t(i)$. The update rule is $w_{t+1}(i) = w_t(i) \cdot \exp(\eta\, \hat{r}_t(i))$.

Using $e^x \le 1 + x + (e-2)x^2$ for $x \le 1$ (valid since $\eta\,\hat{r}_t(i) \le \eta/p_t(i) \le \eta K/\gamma = 1$):

$$\begin{aligned}\frac{\Phi_{t+1}}{\Phi_t} &= \sum_{i=1}^{K} \frac{w_t(i)}{\Phi_t}\, e^{\eta \hat{r}_t(i)} \\ &\le \sum_{i=1}^{K} p'_t(i)\left(1 + \eta \hat{r}_t(i) + (e-2)\eta^2 \hat{r}_t(i)^2\right) \\ &= 1 + \eta \sum_{i=1}^{K} p'_t(i)\hat{r}_t(i) + (e-2)\eta^2 \sum_{i=1}^{K} p'_t(i)\hat{r}_t(i)^2.\end{aligned} \tag{24}$$

Now $p'_t(i) \le p_t(i)/(1-\gamma)$, so $p'_t(i)\hat{r}_t(i)^2 \le p_t(i)\hat{r}_t(i)^2/(1-\gamma) = r_t(I_t)\hat{r}_t(i)/(1-\gamma)$. Since $\hat{r}_t(i) \ge 0$, we take log of both sides, apply $\log(1+x) \le x$, sum over $t = 1, \dots, T$, and take expectations:

$$\begin{aligned}\mathbb{E}[\log \Phi_{T+1}] \le \log K \;+\; &\frac{\eta}{1-\gamma} \sum_{t=1}^{T} \mathbb{E}\big[\textstyle\sum_i p'_t(i)\hat{r}_t(i)\big] \\ &+\; \frac{(e-2)\eta^2}{1-\gamma} \sum_{t=1}^{T} \mathbb{E}\big[r_t(I_t) \textstyle\sum_i \hat{r}_t(i)\big].\end{aligned} \tag{25}$$

c) Step 2: Lower bound via best arm.: For any fixed arm j,

$$\log \Phi_{T+1} \ge \log w_{T+1}(j) = \eta \sum_{t=1}^{T} \hat{r}_t(j).$$

Taking expectations and noting $\mathbb{E}[\hat{r}_t(j)] = r_t(j)$:

$$\mathbb{E}[\log \Phi_{T+1}] \ge \eta \sum_{t=1}^{T} r_t(j).$$

Since this holds for every j, it holds for $j^* = \arg\max_j \sum_t r_t(j)$.

d) Step 3: Assembling the bound.: Combining the upper and lower bounds on $\mathbb{E}[\log \Phi_{T+1}]$ and simplifying the estimates $\mathbb{E}[\sum_i p'_t(i)\hat{r}_t(i)] \le \mathbb{E}[r_t(I_t)]/(1-\gamma)$ and $\mathbb{E}[r_t(I_t)\sum_i \hat{r}_t(i)] \le K$:

$$\eta \sum_{t=1}^{T} r_t(j^*) \le \log K + \frac{\eta}{1-\gamma} \sum_{t=1}^{T} \frac{\mathbb{E}[r_t(I_t)]}{1-\gamma} + \frac{(e-2)\eta^2 K T}{1-\gamma}. \tag{26}$$

Let $G_{\max} = \sum_{t=1}^{T} r_t(j^*)$ and $G_{\text{EXP3}} = \mathbb{E}[\sum_{t=1}^{T} r_t(I_t)]$. Rearranging, with $\eta = \gamma/K$ and $\gamma \le 1/2$ (so $1/(1-\gamma)^2 \le 4$):

$$\begin{aligned}G_{\max} - G_{\text{EXP3}} &\le \frac{K \log K}{\gamma} + (e-2)\gamma T + \frac{4\gamma}{K} G_{\text{EXP3}} \\ &\le \frac{K \log K}{\gamma} + (e-2)\gamma T + 4\gamma T,\end{aligned} \tag{27}$$

since $G_{\text{EXP3}} \le KT$ (each round's reward bounded by K).

e) Step 4: Optimal tuning.: Choose $\gamma = \min\{1, \sqrt{K \ln K/((e-1)T)}\}$. For $T \ge K \ln K/(e-1)$, this gives $\gamma \le 1$ and:

$$\begin{aligned}G_{\max} - G_{\text{EXP3}} &\le \frac{K \log K}{\sqrt{K \ln K/((e-1)T)}} + (e-1)\sqrt{\frac{K \ln K}{(e-1)T}} \cdot T \\ &= 2\sqrt{(e-1)KT \ln K}.\end{aligned} \tag{28}$$

The additive $O(\ln K)$ term in the theorem statement arises from the edge case $T < K \ln K/(e-1)$ where $\gamma = 1$ and the regret is trivially bounded by $KT + \ln K \le 2K \ln K$. □

B. Full Rule Set

The complete set of rules in the NEUROPROOF calculus (38 rules):

Axiom-like: AXIOM, TRUTH.

ND Introduction: AND_I, OR_I_L, OR_I_R, IMP_I, NOT_I, IFF_I.

ND Elimination: AND_E_L, AND_E_R, OR_E, IMP_E (modus ponens), NOT_E, BOT_E (ex falso), DNE (double negation), IFF_E_L.

Sequent Calculus: SC_AX, SC_CUT, SC_WEAKL, SC_WEAKR, SC_AND_R, SC_IMP_R.

Resolution: RES_UNIT, RES_FULL, RES_FACTOR.

NeuroProof novel rules: ADAPTIVE_CUT (theorem 3.8), INTERPOLANT (section VI), LEMMA_REUSE (section VI).